

Threats, Vulnerabilities and Controls

Foundation Concepts of Information Security

Introduction

Threats, Vulnerabilities and Controls form the foundation of information security.

Understanding these three concepts helps organizations identify risks and implement proper safeguards to protect their information assets.

Threat

Definition:

A threat is any event or action that can cause harm to information systems.

Threats may be:

- Intentional
- Unintentional
- Natural

Examples of Threats

1. Malware
2. Hackers
3. Insider attacks
4. Power failures
5. Floods
6. Earthquakes

These threats can damage systems, steal data, or disrupt operations.

Threat Overview

Threat

- Hacker
- Virus
- Insider Misuse
- Flood
- Fire

Threats can come from both human and natural sources.

Vulnerability

Definition:

A vulnerability is a weakness that can be exploited by a threat.

Examples:

- Weak passwords
- Outdated software
- Unpatched systems
- Poor access controls
- Misconfigured servers

Vulnerability Scenario

Threat: Hacker

Vulnerability: Weak password

Result: Unauthorized system access

This example shows how a small weakness can lead to a serious security breach.

Control

Definition:

A control is a safeguard or countermeasure used to reduce risk.

Controls help prevent, detect, or correct security incidents.

Types of Controls

- **Preventive Controls** Prevent attacks before they occur. Examples: Firewalls, Access control systems, Encryption
- **Detective Controls** Identify attacks during or after occurrence. Examples: Intrusion Detection Systems, Security Logs, Monitoring Systems
- **Corrective Controls** Restore systems after an attack. Examples: Backups, Recovery Systems, Incident Response Plans

Relationship Between Threat, Vulnerability and Control

- Threat leads to Vulnerability
- Vulnerability creates Risk
- Control Applied reduces Risk
- **Example:**
- Threat: Cybercriminal
- Vulnerability: Weak Password
- Control: Multi-Factor Authentication
- Result: Reduced Risk

Summary

- Threat is any event that can cause harm
- Vulnerability is a weakness that can be exploited
- Control is a safeguard used to reduce risk
- These three concepts work together in information security
- Proper understanding helps organizations build stronger protection

Thank You

Questions?